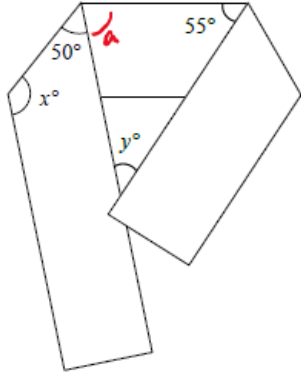
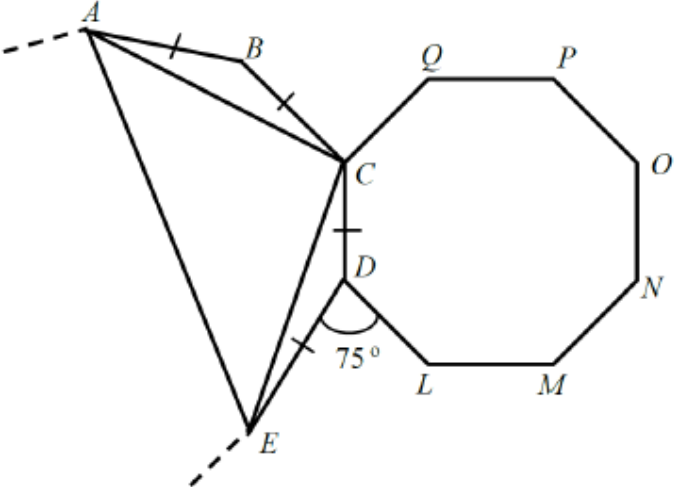
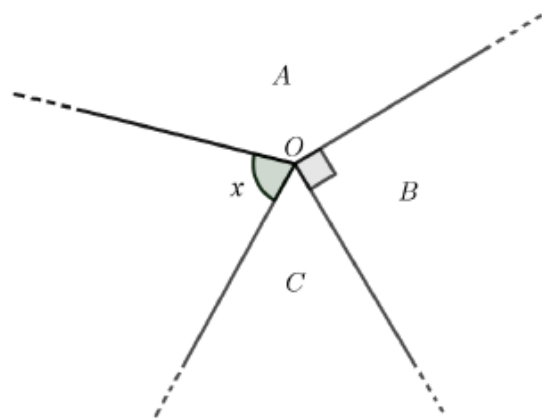


1	YTSS/I/2025/16
	The figure below is a rectangular strip of paper folded at 2 positions.  By giving a reason for each step of your working, find the value of (a) x, [1] (b) y [2]
2	YTSS/I/2025/25
	The diagram shows regular octagon $CDLMNOPQ$ and part of a regular n-sided polygon, $ABCDE \dots$.  (a) Find angle CDE. [2] (b) Find the value of n. [2] (c) Is it possible to have a circle with diameter AE and point C on its circumference? Explain. [2]

- (a) A polygon has n sides. Two of its exterior angles are 50° and 70° , and the remaining exterior angles are each 16° . Find the value of n .

[2]

- (b)

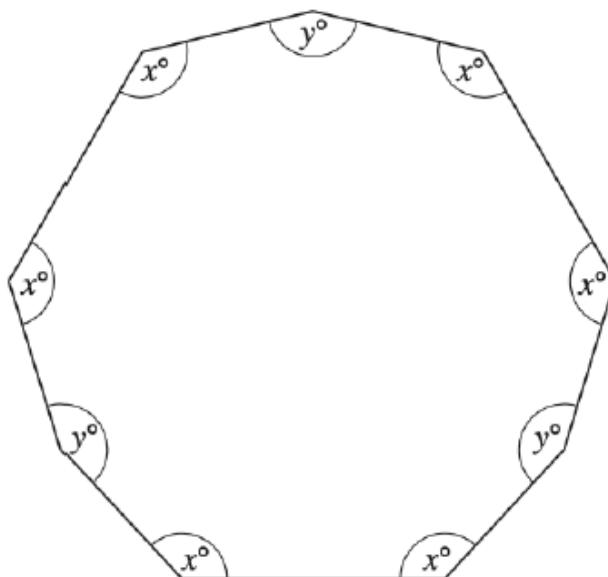


The diagram shows one interior angle of each of three regular polygons, A , B and C . The polygons fit together at point O , leaving a space marked by angle x . A is a regular nonagon, B is a square and C is an equilateral triangle. Find angle x .

[2]

The diagram shows a nonagon.

Six interior angles are each x° . The remaining three interior angles are each y° .



(a) State the number of lines of symmetry if $x^\circ \neq y^\circ$.

[1]

(b) Find the sum of the interior angles of the nonagon.

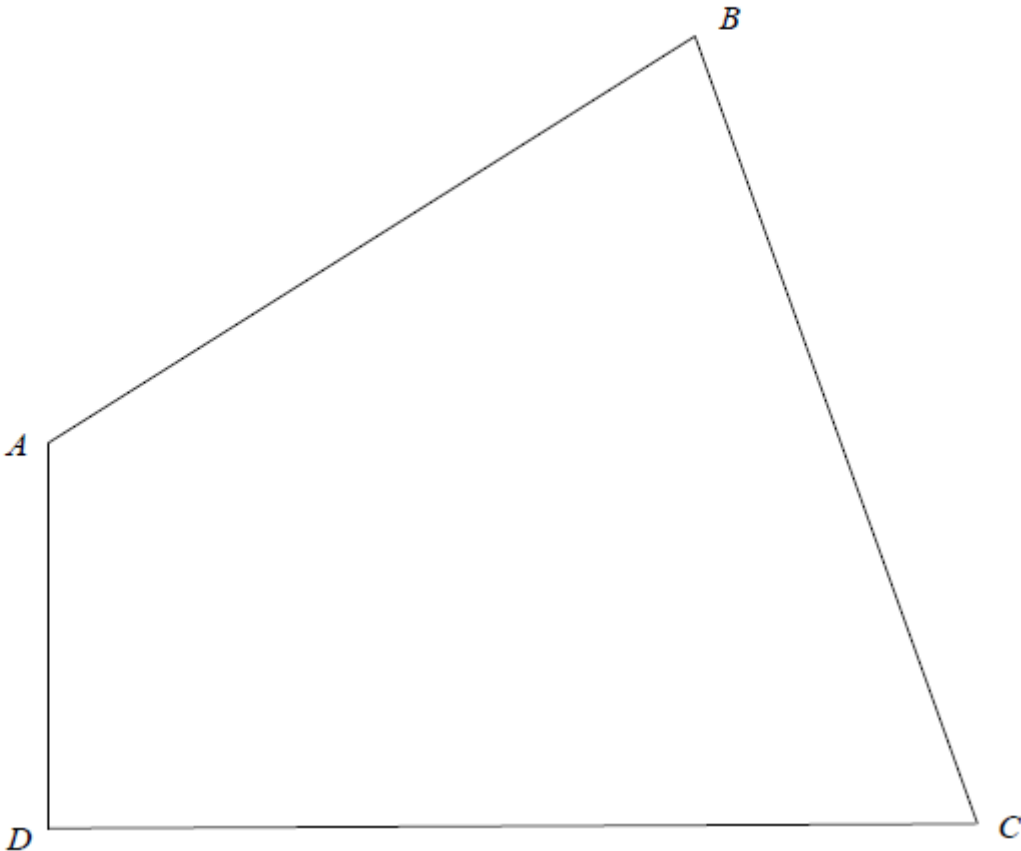
[1]

(c) Hence, show that $y = -2x + 420$.

[1]

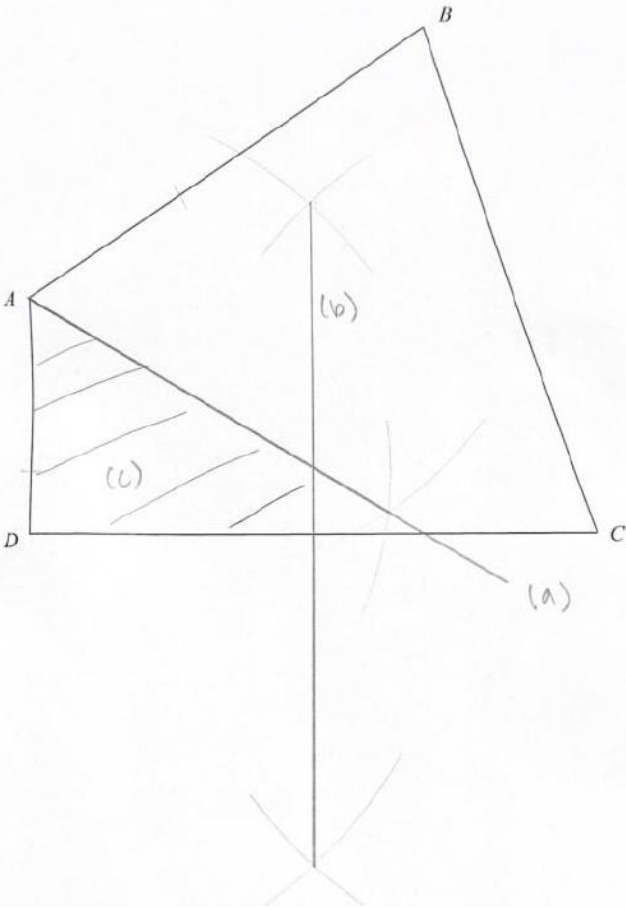
(d) Given that $x : y = 13 : 16$, find x .

[2]

5	GMSS/2025/I/14
	A polygon has exterior angles $(x - 30)^\circ$, $2x^\circ$, $(x + 10)^\circ$, $(x + 45)^\circ$ and $(2x - 15)^\circ$. (a) Find x. [2] (b) Find the smallest interior angle of the polygon. [1]
6	ACSB/2025/I/10
	The diagram represents a plot of land, $ABCD$, set aside as nature reserve.  (a) Construct the bisector of angle DAB. [1] (b) Construct the perpendicular bisector of DC. [1] (c) An animal shelter is to be built in the nature reserve, nearer to D than to C and nearer to AD than to AB. Shade the region where the shelter is to be built. [1]

7	ACSBR/2025/I/14
	In a regular polygon, the ratio of the interior angle : exterior angle = 5 : 1 . Calculate the number of sides of the polygon. [2]

1	(a) $x = 180 - 50 = 130$ (int angles, // lines) --- [B1] No marks awarded if no reason (b) $a = 180 - 50 - 50 = 80$ (adj angles on a st line) --- [M1] $y = 180 - 80 - 55 = 45^\circ$ (sum of angles in a triangle) --- [A1] Minus 1 mark if no reason
2	(a) $\angle CDL = \frac{(8-2) \times 180^\circ}{8} = 135^\circ$ --- [M1] $\angle CDE = 360 - 135 - 75 = 150^\circ$ (angles at a point) --- [A1] (b) ext angle = $180 - 150 = 30$ (adj angles on a st line) --- [M1] $30n = 360$ $n = 12$ --- [A1] <u>Alternative:</u> $(n-2)(180) = 150n$ $30n = 360$ $n = 12$ (c) $\angle DCE = \frac{180-150}{2} = 15^\circ$ ---- [M1] $\angle ACE = 150 - 2(15) = 120^\circ$ Since $\angle ACE = 120^\circ \neq 90^\circ$, by the converse of property angles in a semicircle, it is not possible to form a circle with diameter AE and point C on circumference. [A1]
3	(a) $50 + 70 + 16(n-2) = 360$ $16n = 272$ $n = 17$ (b) $x = 360 - \left(\frac{(9-2) \times 180}{9} + 90 + 60 \right)$ $= 70^\circ$
4	(a) Lines of symmetry = 3 (b) $(9-2) \times 180^\circ = 1260^\circ$ (c) Use $6x^\circ + 3y^\circ = 1260^\circ$ $y = -2x + 420$ (shown) (d)

	$\frac{x}{y} = \frac{13}{16}$ $y = \frac{16}{13}x \text{ --- (1)}$ Sub (1) into equation into $y = -2x + 420$ $-2x + 420 = \frac{16}{13}x$ $-26x + 5460 = 16x$ $42x = 5460$ $x = 130$
5	(a) $(x - 30) + 2x + (x + 10) + (x + 45) + (2x - 15) = 360$ $7x + 10 = 360$ $7x = 350$ $x = 50$ (b) Largest exterior angle = $2(50^\circ) = 100^\circ$ Smallest interior angle = $180^\circ - 100^\circ = 80^\circ$
6	
7	$\frac{180}{6} = 30$ $\frac{360}{30} = 12$